

# Sufficient Statistics

2026-04-17

## Introduction

A statistic is **sufficient** for a parameter if it captures all the information in the data that is relevant to that parameter. Once a sufficient statistic is computed, the raw data have nothing left to add about the parameter. Sufficiency organises the theory of estimation: it explains why the sample mean is a complete summary for the normal mean, why the sum of successes suffices for the binomial proportion, and why the exponential family of distributions is central to modern statistical modelling.

## Prerequisites

The reader should be comfortable with the likelihood function, conditional probability, and a little abstract notation.

## Theory

Formally,  $T(X_1, \dots, X_n)$  is **sufficient for  $\theta$**  if the conditional distribution of the data given  $T = t$  does not depend on  $\theta$ . Equivalently (Fisher-Neyman factorisation theorem),  $T$  is sufficient iff the joint density factorises as

$$f(x_1, \dots, x_n \mid \theta) = g(T(x), \theta) \cdot h(x),$$

where  $h$  does not depend on  $\theta$  and  $g$  depends on the data only through  $T$ .

Examples:

- **Normal** ( $\mu, \sigma^2$ ) **with known**  $\sigma^2$ :  $T = \bar{X}$  is sufficient for  $\mu$ .
- **Normal with both unknown**:  $(\bar{X}, \sum (X_i - \bar{X})^2)$  is jointly sufficient.
- **Bernoulli/Binomial**:  $T = \sum X_i$  is sufficient for  $\pi$ .
- **Poisson**:  $T = \sum X_i$  is sufficient for  $\lambda$ .
- **Uniform**( $0, \theta$ ):  $T = \max X_i$  is sufficient for  $\theta$ .

A statistic is **minimal sufficient** if it is a function of every other sufficient statistic – i.e., it is the most compact sufficient summary.

The **exponential family** has density  $f(x \mid \theta) = h(x) \exp[\eta(\theta)^\top T(x) - A(\theta)]$  for natural statistic  $T(x)$ . Almost every distribution used in applied statistics (normal, binomial, Poisson, gamma, beta, multinomial) belongs to the exponential family, and  $T(x)$  is minimal sufficient for  $\eta(\theta)$ .

**Rao-Blackwell theorem**: given any unbiased estimator  $\hat{\theta}$  and a sufficient statistic  $T$ , the conditional expectation  $E[\hat{\theta} \mid T]$  is also unbiased and has variance no larger than  $\hat{\theta}$ . Conditioning on sufficient statistics never hurts and may help.

## Assumptions

The factorisation theorem applies to proper densities on a common support. In non-regular families (uniform with an unknown upper endpoint) sufficiency still applies, but the resulting estimators inherit the non-standard behaviour of the family.

## R Implementation

```
set.seed(2026)

n <- 50
mu <- 5
sigma <- 2
x <- rnorm(n, mean = mu, sd = sigma)

T1 <- sum(x)
T2 <- sum(x^2)

log_lik_from_sum <- function(mu, sigma, T1, T2, n) {
  -n/2 * log(2 * pi * sigma^2) -
    (T2 - 2 * mu * T1 + n * mu^2) / (2 * sigma^2)
}

mu_hat <- T1 / n
sigma2_hat <- (T2 - T1^2/n) / n

c(mu_hat = mu_hat, sigma2_hat = sigma2_hat)

c(mle_from_data_mean = mean(x),
  mle_from_data_var = mean((x - mean(x))^2))
```

Computing the MLE from only the two sufficient statistics  $(T_1, T_2) = (\sum x, \sum x^2)$  yields the same point estimates as computing it from the full data vector.

## Output & Results

```
mu_hat sigma2_hat
5.090      3.821

mle_from_data_mean mle_from_data_var
5.090              3.821
```

The two are numerically identical, confirming that  $(T_1, T_2)$  contains all the information about the parameters that the data can provide.

## Interpretation

In practice, sufficiency is used under the hood. When a software package reports a confidence interval for a normal mean using `t.test()`, it relies on the fact that  $\bar{X}$  and  $s^2$  are sufficient for  $(\mu, \sigma^2)$ . For a manuscript, sufficiency is rarely mentioned explicitly but justifies the use of summary statistics as complete descriptions of the data in the parametric model assumed.

## Practical Tips

- Sufficient statistics let you store compressed summaries (counts, sums) without losing information for inference – useful for distributed computation.
- The exponential family is the most important family to understand, because almost every GLM assumes it.
- Completeness plus sufficiency plus unbiasedness implies the UMVUE (Lehmann-Scheffe); many textbook UMVUE proofs rely on this.

- Rao-Blackwellisation is useful in MCMC: conditioning on sufficient statistics for part of the parameter vector reduces Monte Carlo variance.
- Sufficiency depends on the model. A statistic that is sufficient for  $\mu$  in a normal model may carry no information about the same  $\mu$  under a different data-generating process.

## For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr